

Zhongkao Geometry Complete Models

(Teaching & Exam Edition)

11 Modules · 47+ Geometry Models · Full Scoring Sprint

Each chapter includes exercise whiteboard pages

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Final: Exam Review Summary

Identify	[Progress bar]
Conclusion	[Progress bar]
Steps	① [Progress bar] ② / [Progress bar] ③ [Progress bar] ④ [Progress bar]
Mnemonic	[Progress bar]
Pitfalls	[Progress bar]

Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____

[illegible]

Exercises: ■■■■■■■■■■■■■■■■

(Exercise content area - fill with problems)

Tip: Enter practice problems above, students solve in this space

Q1: _____

Q2: _____

Q3: _____

Identify

Conclusion

Steps

Mnemonic

Pitfalls

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①=②③

①②③

5. +

Essence

Identify

Conclusion

Steps

Mnemonic

Pitfalls

AM=BN

①②③

①②③④

6.

Essence

Identify

Conclusion

Steps

Mnemonic

Pitfalls

180

Quadrilateral with supplementary diagonal and one pair of equal adjacent sides.

1. Rotate to congruent triangles. 2. Fixed triangle area.

1. Check supplementary and equal sides. 2. Rotate. 3. Substitute.

Supplementary diagonals with equal sides.

Missing equal adjacent side condition.

Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____



1. ■■■■■■

Essence	
Identify	Triangle midpoint/median with scattered segments.
Conclusion	Doubled median yields congruent triangles with equal sides and angles.
Steps	1. Find triangle median. 2. Extend to double length. 3. Connect endpoint. 4. Transfer.
Mnemonic	See median, double it.
Pitfalls	Insufficient extension length.



2. ■■■■■■■■■■

Essence	
Identify	Right triangle with hypotenuse midpoint.
Conclusion	Hypotenuse median = half hypotenuse, forms isosceles.
Steps	1. Lock right triangle and midpoint. 2. Connect median. 3. Use equal angles.
Mnemonic	Right triangle midpoint, half gives isosceles.
Pitfalls	Using on non-right triangles.



3. ■■■■■■

Essence	
Identify	Two midpoints in triangle.
Conclusion	Mid-segment parallel to base, length = half base.
Steps	1. Identify two midpoints. 2. Connect. 3. Apply parallel and half-length.
Mnemonic	Two midpoints connect, parallel and half.
Pitfalls	Single midpoint mistaken for mid-segment.

Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____



1. ■■■■■■■■■■

Essence	
Identify	Shared angle, inner triangle, right triangle altitude.
Conclusion	1. Large similar to small. 2. Projection theorem: leg squared = hypotenuse segment x hypotenuse.
Steps	1. Find shared and equal angles. 2. Identify similarity. 3. Set proportions.
Mnemonic	Shared angle similarity, square product directly known.
Pitfalls	Wrong vertex correspondence.



2. ■■■■■■■■■■

Essence	
Identify	Multiple parallel lines crossing two lines.
Conclusion	Corresponding segments proportional.
Steps	1. Identify parallel group. 2. Set proportions. 3. Substitute.
Mnemonic	Parallel lines cut segments proportionally.
Pitfalls	Wrong segment correspondence.

Exercises: ■■■■■■■■■■■■■■■

(Exercise content area - fill with problems)

Tip: Enter practice problems above, students solve in this space

Q1: _____

Q2: _____

Q3: _____



Essence	
Identify	Two fixed points, one moving point on line, find min/max sum or perimeter.
Conclusion	1. Min sum: reflect, connect, intersection. 2. Max diff: extend line.
Steps	1. Reflect point across line. 2. Connect to other point. 3. Intersection is optimal.
Mnemonic	Min sum reflect, max diff connect directly.
Pitfalls	Wrong reflection point.



Essence	 $PA + k \cdot PB$
Identify	Moving point on line, find min of $PA + k \cdot PB$.
Conclusion	After angle construction, $k \cdot PB$ becomes perpendicular, min = perpendicular distance.
Steps	1. Identify $PA + k \cdot PB$. 2. Construct special angle from k. 3. Convert to perpendicular. 4. Solve.
Mnemonic	Line moving with coefficient, construct angle for shortest.
Pitfalls	Wrong angle construction.



Essence	 PA+k*PB
Identify	Moving on circle, coefficient k, find PA+k*PB extreme.
Conclusion	After interior similarity, convert to standard optimization.
Steps	1. Identify circle, fixed points, moving point. 2. Construct similarity via radius. 3. Scale. 4. Solve.
Mnemonic	Circle with coefficient, center similarity to convert.
Pitfalls	Confused with Hu Bu Gui (line vs circle).



Essence 60

Mnemonic

Fixed product inversion, line-circle transformation.

Pitfalls

Confused with melon-seed.

Exercises: ■■■■■■■■■■&■■■

(Exercise content area - fill with problems)

Tip: Enter practice problems above, students solve in this space

Q1: _____

Q2: _____

Q3: _____



1. ■■■■

Essence	
Identify	Shape folded along a line.
Conclusion	Corresponding sides and angles equal. Fold line perpendicularly bisects connecting points.
Steps	1. Mark equal parts. 2. Set variables. 3. Use Pythagorean or similarity equations.
Mnemonic	Fold = symmetry, equal sides and angles.
Pitfalls	Missing hidden equal parts.



2. ■■■■■■

Essence	
Identify	Fixed segment sliding on perpendicular axes/walls.
Conclusion	Midpoint locus = quarter circle arc. Distance from midpoint to fixed point constant.
Steps	1. Identify fixed-length sliding. 2. Use right triangle midpoint property. 3. Find circle center and radius.
Mnemonic	Fixed-length perpendicular slide, midpoint circular arc.
Pitfalls	Cannot identify circular locus.

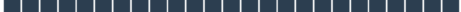
Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____



1. ■■■■■■

Essence	
Identify	Polygon in circle, diameter, cyclic quadrilateral.
Conclusion	1. Same arc equal angles. 2. Diameter gives right angle. 3. Cyclic supplementary sum 180.
Steps	1. Find same arc, diameter, cyclic quad. 2. Apply angle properties. 3. Combine with congruence.
Mnemonic	Same arc equal, diameter right, cyclic supplementary.
Pitfalls	Confusing same arc vs equal arc.



2. ■■■■

Essence	
Identify	Circle tangent, tangent point, radius.
Conclusion	1. Tangent perpendicular to radius. 2. Tangent-chord angle = inscribed angle of intercepted arc.
Steps	1. Prove tangent: connect radius, prove perpendicular. 2. Use perpendicular and tangent-chord angle.
Mnemonic	Tangent perpendicular to radius.
Pitfalls	Non-standard proof steps.

Exercises:

(Exercise content area - fill with problems)

Tip: Enter practice problems above, students solve in this space

Q1: _____

Q2: _____

Q3: _____

Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____

Essence



10. ■■■■■■■■■■■■■■■■■■■■

Essence



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(Exercise content area - fill with problems)

Tip: Enter practice problems above, students solve in this space

Q1: _____

Q2: _____

Q3: _____



1. ■■■■■■

Essence	
Identify	Quadrilateral with perpendicular diagonals.
Conclusion	$AB^2 + CD^2 = AD^2 + BC^2$
Steps	1. Check perpendicular diagonals. 2. Apply formula. 3. Substitute.
Mnemonic	Vertical diagonals, opposite square sums equal.
Pitfalls	Applying to non-vertical quadrilaterals.



2. ■■■■■■

Essence	
Identify	Any quadrilateral, connect all side midpoints.
Conclusion	1. Always parallelogram. 2. Equal diagonals->rhombus. 3. Perpendicular->rectangle. 4. Both->square. 5. Area=half.
Steps	1. Check original diagonals. 2. Determine midpoint shape. 3. Apply.
Mnemonic	Midpoints connect, always parallelogram.
Pitfalls	Cannot relate to diagonal properties.



3. ■■■■■■■■■■

Essence	
Identify	Shortest path on box/cylinder/prism surface.
Conclusion	Unfold then straight line. Box: compare 3 unfoldings.
Steps	1. Unfold 3D surface. 2. Right triangle. 3. Pythagorean hypotenuse.
Mnemonic	3D shortest: unfold first.
Pitfalls	Wrong unfolding.



4. $\square\square\square\square\square\square\square\square\square + \square\square\square + \square\square\square$

Essence

Identify	Intersecting chords, secants, tangent+secant.
Conclusion	1. Chords: $PA \cdot PB = PC \cdot PD$. 2. Tangent: $PT^2 = PA \cdot PB$. 3. Secants: $PA \cdot PB = PC \cdot PD$.
Steps	1. Identify type. 2. Apply formula. 3. Solve equation.
Mnemonic	Inside chords product equal. Outside tangent square equal.
Pitfalls	Confusing three formulas.

Exercises: ■■■■■■■■■■

Tip: Enter practice problems above, students solve in this space

Q3: _____

Final: Exam Review Summary

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